

Title

Conversational AI Support for Self-Management of Atrial Fibrillation: Results from a Randomised Trial

Abstract

A trial randomised 480 adults with atrial fibrillation to receive either digital conversational coaching via an interactive platform or standard clinical follow-up. The conversational AI group received access to 24/7 dialogue-based education regarding symptom recognition, medication adherence, and lifestyle modifications relevant to arrhythmia management. Quality-of-life measures specific to atrial fibrillation-related limitations were assessed at trial conclusion. Participants receiving conversational AI support demonstrated numerically higher satisfaction with self-management resources and reported greater confidence in recognising warning signs. User surveys indicated the interactive platform was perceived as accessible and supportive by the majority. Engagement metrics showed sustained interaction with the conversational system over the study period. The usual care cohort completed standard clinic-based education and support. Quantitative quality-of-life scores showed modest improvements in both arms, with the AI-supported group trending toward greater gain, though formal statistical comparison revealed mixed outcomes. This trial provides preliminary evidence that conversational AI may serve as an acceptable adjunct to standard atrial fibrillation care delivery.